CS3002E Compiler Design

Monsoon 2025

Lecture #9

Top-Down Parsing-2

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Predictive Parsing

$$
\begin{aligned}
S &\rightarrow (A) \\
A &\rightarrow C \mid D \\
C &\rightarrow ab \\
D &\rightarrow b
\end{aligned}
$$

Does look-ahead help in choosing the production for A?

Predictive Parsing

$S \rightarrow (A)$
$A \rightarrow C \mid D$
$C \rightarrow ab$
$D \rightarrow b$

Choosing the production for A based on look-ahead:

- if lookahead is a, choose $A \rightarrow C$
- if lookahead is b, choose $A \rightarrow D$

The choice is based on the strings derivable from C and D, more specifically based on the terminal symbols that can begin strings derivable from C and D.

FIRST set: Definition

Define **FIRST**(α), where α is any string of grammar symbols, to be the set of terminals that begin strings derivable from α. If $\alpha \overset{*}{\Rightarrow} \epsilon$, then ϵ is also in $FIRST(\alpha)$.

FIRST sets: example

$\text{FIRST}(ab) = \{a\}$
$\text{FIRST}(b) = \{b\}$

Computing FIRST

$A \to \alpha \mid \beta$

α begins with a terminal a (α is $a\alpha'$)

β begins with a terminal b (β is $b\beta'$)

$a \in FIRST(\alpha)$
$b \in FIRST(\beta)$

FIRST: example

$S \rightarrow aAb$

$FIRST(S) = FIRST(aAb) = \{a\}$

FIRST: example

$$S \rightarrow aAb$$
$$A \rightarrow B \mid C$$
$$B \rightarrow bd$$
$$C \rightarrow c$$

$FIRST(A) = ?$

FIRST: example

$$
\begin{aligned}
S &\rightarrow aAb \\
A &\rightarrow B \ | \ C \\
B &\rightarrow bd \\
C &\rightarrow c
\end{aligned}
$$

$$FIRST(A) = FIRST(B) \cup FIRST(C) = \{b,\ c\}$$

Predictive Parsing: decision based on FIRST sets

$A \rightarrow \alpha \mid \beta$

if lookahead $\in$ $FIRST(\alpha)$ choose $A \rightarrow \alpha$

if lookahead $\in$ $FIRST(\beta)$ choose $A \rightarrow \beta$

Predictive Parsing: FIRST sets are not disjoint

$A \rightarrow \alpha \mid \beta$

if lookahead $\in FIRST(\alpha)$ choose $A \rightarrow \alpha$

if lookahead $\in FIRST(\beta)$ choose $A \rightarrow \beta$

But both $FIRST(\alpha)$ and $FIRST(\beta)$ can contain the same symbols as in the following productions:

A$\rightarrow$ aB | aC

Predictive Parsing based on FIRST sets

$A \rightarrow \alpha \mid \beta$

If $FIRST(\alpha)$ and $FIRST(\beta)$ are disjoint, a parsing decision can be made based on the lookahead symbol.

Computing FIRST[1]

- For a terminal a, $FIRST(a) = \{a\}$
- For a string α of the form $a\alpha'$, $FIRST(\alpha) = \{a\}$
- For a Nonterminal A, for every production $A \rightarrow \alpha$, add every symbol in $FIRST(\alpha)$ to $FIRST(A)$

[1] assuming no ϵ productions

Computing FIRST

S → aAb
A → B | C
B → bd
C → c

$$FIRST(S) = FIRST(aAb) = \{a\}$$
$$FIRST(B) = \{b\}$$
$$FIRST(C) = \{c\}$$
$$FIRST(A) = FIRST(B) \cup FIRST(C) = \{b,\ c\}$$

Predictive Parsing: Exercise

Given the grammar:

$$
\begin{array}{lcl}
\textit{stmt} & \rightarrow & \textbf{id}{=}\textit{expr} \\
 & | & \textbf{if}\ (\textbf{id})\ \textit{stmt} \\
 & | & \textbf{while}\ (\textbf{id})\ \textit{stmt} \\
 & & \\
\textit{expr} & \rightarrow & \textbf{id} \\
 & | & \textbf{num}
\end{array}
$$

Compute FIRST set for each string that appears as the body of a production. Show how these sets help the parser in making parsing decisions for the following inputs.

1. `sum=x1`
2. `if (x) length=0`

Predictive Parsing Table entries based on FIRST sets

Grammar:

$$\begin{aligned} S &\rightarrow \textbf{id} = E \\ E &\rightarrow \textbf{id}+\textbf{id} \mid \textbf{num} \mid -\textbf{id} \end{aligned}$$

$$\begin{aligned} FIRST(id = E) &= \{\text{id}\} \\ FIRST(id + id) &= \{\text{id}\} \\ FIRST(num) &= \{\text{num}\} \\ FIRST(-id) &= \{-\} \\ FIRST(S) &= \{\text{id}\} \\ FIRST(E) &= \{\text{id, num, -}\} \end{aligned}$$

Predictive Parsing Table entries based on FIRST sets

	id	num	$-$	\$
S	$S \to id = E$	*error*	*error*	*error*
E	$E \to id + id$	$E \to num$	$E \to -id$	*error*

for each production $A \to \alpha$
  for each terminal $a \in FIRST(\alpha)$
    add $A \to \alpha$ in entry $T[A,\ a]$

- \$ is the input end marker symbol (not a grammar symbol)
  input: $w\$$
  first sentential form: $S\$$
  $S\$ \stackrel{*}{\Rightarrow} w\$$

Computing FIRST

$$S \rightarrow aAb$$
$$A \rightarrow B \;|\; C$$
$$B \rightarrow bd$$
$$C \rightarrow c$$

$$FIRST(S) = FIRST(aAb) = \{a\}$$
$$FIRST(B) = \{b\}$$
$$FIRST(C) = \{c\}$$
$$FIRST(A) = FIRST(B) \cup FIRST(C) = \{b,\; c\}$$

Predictive Parsing Table

	a	b	c	d	\$
S	$S \rightarrow aAb$				
A		$A \rightarrow B$	$A \rightarrow C$		
B		$B \rightarrow bd$			
C			$C \rightarrow c$		

for each production $A \rightarrow \alpha$
for each terminal $a \in FIRST(\alpha)$
add $A \rightarrow \alpha$ in entry $T[A,\ a]$

Note: *error* entries are left blank

Predictive Parsing Table

$$
\begin{aligned}
S &\rightarrow\ (A) \mid [A] \\
A &\rightarrow\ \mathbf{id} + E \mid \mathbf{num} + E \\
E &\rightarrow\ \mathbf{id} \mid \mathbf{num}
\end{aligned}
$$

	(	[	id	num	)	]	$
S	$S \rightarrow (A)$	$S \rightarrow [A]$					
A			$A \rightarrow id + E$	$A \rightarrow num + E$			
E			$E \rightarrow id$	$E \rightarrow num$			

Exercise

Build a Predictive Parsing Table for the grammar given below:

$$S \rightarrow \ \mathbf{id} = E \mid \mathbf{if}\ (E)\ \mathbf{id}$$
$$E \rightarrow \ \mathbf{id} + \mathbf{id} \mid \mathbf{num} \mid \ - \mathbf{id}$$

Computing FIRST - CFG with ϵ productions

$$
\begin{aligned}
S &\rightarrow (A)\\
A &\rightarrow Cd\\
C &\rightarrow ab \mid \epsilon
\end{aligned}
$$

$A \stackrel{*}{\Rightarrow} Cd \stackrel{*}{\Rightarrow} d$

$d \in FIRST(Cd)$

$d \in FIRST(A)$

$$
\begin{aligned}
FIRST(S) &= \{\ (\ \}\\
FIRST(A) &= \{a,\ d\}\\
FIRST(C) &= \{a,\ \epsilon\}
\end{aligned}
$$

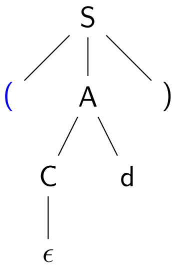
If $\alpha \stackrel{*}{\Rightarrow} \epsilon$, add ϵ to $FIRST(\alpha)$.

CFG with ϵ productions

$$A \rightarrow Cd$$
$$C \rightarrow ab \mid \epsilon$$

- In which cell of the parsing table, the entry for $C \rightarrow \epsilon$ is to be made?
- What should be the look-ahead in order to apply $C \rightarrow \epsilon$?

Top-Down Parsing - choosing a production



Input: ($_{\uparrow}$d)

- after matching '(' look-ahead is d, not a symbol in FIRST(C)
- but d can appear next to C (or follow C) in a sentential form
- choose $C \rightarrow \epsilon$, when the look-ahead is d
  - may result in a match in the next step

FOLLOW

Define $FOLLOW(A)$, for nonterminal A, to be the set of terminals a that can appear immediately to the right of A in some sentential form; that is the set of terminals a such that there exists a derivation $S \stackrel{*}{\Rightarrow} \alpha A a \beta$, for some α and β. In addition, if A can be the rightmost symbol in some sentential form, then \$ is in $FOLLOW(A)$[2].

[2] \$ is the special input "endmarker" symbol that is assumed to be not in the grammar.

FOLLOW

$$S \to (A)$$
$$A \to Cd$$
$$C \to ab \mid \epsilon$$

$$FOLLOW(A) = \{\ )\ \}$$
$$FOLLOW(C) = \{\ d\ \}$$
$$FOLLOW(S) = \{\ \$\ \}$$

Input: $w\$$, first sentential form: $S\$$

$$S\$ \overset{*}{\Rightarrow} w\$$$

Parsing table entries based on FOLLOW sets

$S \rightarrow (A)$
$A \rightarrow Cd$
$C \rightarrow ab \mid \epsilon$

	(	)	a	b	d	\$
S	$S \rightarrow (A)$					
A			$A \rightarrow Cd$		$A \rightarrow Cd$	
C			$C \rightarrow ab$		$C \rightarrow \epsilon$	

for each production $A \rightarrow \alpha$
  if $\epsilon \in FIRST(\alpha)$
    for each terminal $a \in FOLLOW(A)$
      add $A \rightarrow \alpha$ in $T[A,\ a]$

Exercise

Build a predictive parsing table for the following grammar:

$$
\begin{array}{lcl}
S & \rightarrow & (E) \\
E & \rightarrow & idA \\
A & \rightarrow & +id \mid \epsilon
\end{array}
$$

Reference:

- Aho A.V., Lam M.S., Sethi R., and Ullman J.D. Compilers: Principles, Techniques, and Tools (ALSU). Pearson Education, 2007.

Further reading:

- ALSU Chapter 4, Sections 4.3.4, 4.4.